

Expert Evaluation Questionnaire — University Surveillance System

A Model-Driven Requirements Engineering Approach for Context-Aware IoT Systems
Applied to the University Surveillance System

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This questionnaire is designed to take approximately 45 minutes. Please refer to the Models provided separately while answering. There are no right or wrong answers. We are interested in your honest assessment, including disagreement.

Reading Guide

Approximate time: 45 minutes total.

Sections in this form:

- Demographic & expertise (4 min)
- A. UCM4IoT use case models (9 min)
- B. Dynamic Goal Modelling — DGM (12 min)
- C. CARDML4IoT dependency models (7 min)
- D. Comparative assessment vs current approach (6 min)
- E. Perceived usefulness, ease of use, intention to use (5 min)
- F. Open-ended overall feedback (2 min)

Likert scale (1–5): 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree. Each Likert item has an 'Anchor' that describes what 'Agree' concretely means for that item; please use it as your reference.

If you genuinely have no opinion or insufficient information to judge, the middle option (3 = Neutral) is the right answer.

Please refer to the Models (provided separately) while answering Sections A, B, and C.

Demographic and Expertise Information

Approximate time: 4 minutes. The information collected here is used only to characterize the panel of reviewers in aggregate.

1. D-1. Current role and years of experience in software / requirements engineering
e.g., 'Researcher, 6 years'

2. D-2. Years of experience working on the University Surveillance System
0 if you joined recently.

3. D-3. Your role on the system
e.g., Designer, Developer, Operator, Researcher; multiple allowed.

4. D-4. Modelling experience (check all that apply)

Check all that apply.

- ☐ i* / iStar or KAOS or GRL
- ☐ BPMN
- ☐ UML use case modelling
- ☐ UML state machines or activity diagrams
- ☐ Event-Condition-Action (ECA) rules
- ☐ Model-driven engineering tools (Ecore, Xtext, EMF, ATL, etc.) or DSLs
- ☐ None of the above

5. D-5. Familiarity with the modelling languages used in this evaluation

Mark the box that best describes your familiarity before reading the Models Booklet.

Mark only one oval per row.

	Never heard of it	Heard of it	Read about it	Used it
UCM4IoT (use case modelling for IoT)	☐	☐	☐	☐
DGM (Dynamic Goal Modelling)	☐	☐	☐	☐
CARDML4IoT (CA dependency modelling)	☐	☐	☐	☐

Section A. UCM4IoT Use Case Models

Supports RQ1 (Requirement Completeness). Approximate time: 9 minutes. Please refer to the UCM4IoT use case description models (Section A) before completing this section.

A.1 Global ratings on the UCM4IoT models

6. A1-1. The UCM4IoT models capture the main functional requirements of the University Surveillance System as it is currently designed and operated.

Anchor: 'Agree' means: at least the major use cases (camera capture, detection, event publication, alert handling, dashboard display) are present and the main interactions in each are correctly described.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

7. A1-2. The UCM4IoT models capture the IoT-specific actor types (sensors, actuators, software systems, physical entities, human users) that participate in the system.

Anchor: 'Agree' means: the models distinguish camera/sensor actors from software systems, and from human operators/agents, in line with how the system actually decomposes responsibilities.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

8. A1-3. The UCM4IoT models capture context-awareness aspects that depend on time of day, location, actor role, or operating mode (e.g., differentiated handling of an event during the day vs. at night).

Anchor: 'Agree' means: contextual conditions of this kind are visible in the models (in CA requirement annotations or extension blocks), not only encoded implicitly in the prose.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

9. A1-4. The UCM4IoT exception and handler use cases reflect the kinds of failure and abnormal situations the system actually has to handle (e.g., camera offline, network unreachable, detection model failure, ambiguous detection).

Anchor: 'Agree' means: the major failure modes you encounter operationally are represented in extension blocks and have associated handler use cases.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

10. A1-5. The UCM4IoT notation, on first reading, is clear enough that I can determine what the system is supposed to do in each use case without external explanation.

Anchor: 'Agree' means: the textual notation is self-readable. 'Disagree' means: I needed to look up syntax frequently or could not tell what a use case meant.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

11. A1-6. The UCM4IoT models are too coarse-grained: they describe the system at a level of abstraction that hides important interactions a designer would need to specify.

Anchor: 'Agree' means: the models miss details that matter for design. 'Disagree' means: the abstraction level is appropriate for a requirements artefact.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

A.2 Per-use-case coverage

For each UCM4IoT use case in the Models (Section A), please rate how completely and accurately it captures the corresponding behaviour as you know it. Note any actor misclassifications (e.g., something modelled as a sensor that should be a software system) in the comments.

12. A2. Captured? (Y / Partial / N)

Y = captured correctly and completely. Partial = partly captured but missing details, has minor inaccuracies, or could be more precise. N = missing or substantially incorrect. Use the Comments box below for specifics.

Mark only one oval per row.

	Y	Partial	N
UC-1 Camera capture and frame publication	☐	☐	☐
UC-2 License plate recognition	☐	☐	☐
UC-3 Plate restriction check (against authorization database)	☐	☐	☐
UC-4 Body pose detection	☐	☐	☐
UC-5 Suspicious behaviour detection (combining pose + plate + spatial + temporal evidence)	☐	☐	☐
UC-6 Weapon detection (firearm / bladed)	☐	☐	☐
UC-7 Fire and smoke detection	☐	☐	☐

UC-8 Flood

UC-8 Flood detection	☐	☐	☐
UC-9 Event hub			
UC-9 Event deduplication, persistence, prioritization	☐	☐	☐
UC-10 Operator dashboard alerting and field agent notification	☐	☐	☐

13. A2. Accuracy of each element (1–5)

1 = very inaccurate, 5 = very accurate. Use 3 if you cannot judge.

Mark only one oval per row.

	1	2	3	4	5
UC-1 Camera capture and frame publication	☐	☐	☐	☐	☐
UC-2 License plate recognition	☐	☐	☐	☐	☐
UC-3 Plate restriction check (against authorization database)	☐	☐	☐	☐	☐
UC-4 Body pose detection	☐	☐	☐	☐	☐
UC-5 Suspicious behaviour detection (combining pose + plate + spatial + temporal evidence)	☐	☐	☐	☐	☐
UC-6 Weapon detection (firearm / bladed)	☐	☐	☐	☐	☐
UC-7 Fire and smoke detection	☐	☐	☐	☐	☐
UC-8 Flood detection	☐	☐	☐	☐	☐

**UC-9 Event
hub:
deduplication,
persistence,
prioritization**

☐	☐	☐	☐	☐
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**UC-10
Operator
dashboard
alerting and
field-agent
notification**

☐	☐	☐	☐	☐
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14. A2 comments. If you marked 'Partial' or 'N' for any use case, or if any rating warrants explanation, please specify here. Reference rows by their ID (e.g., 'UC-3: ...').

15. A2-1. Are there functional requirements of the system that are not covered by any of the listed use cases? If yes, please list them, naming the actors and the main interaction.

16. A2-2. Did the UCM4IoT models reveal any requirement that you had not previously considered or documented? If yes, please describe.

Section B. Dynamic Goal Modelling (DGM)

Supports RQ2 (Dynamic Goal Coverage). Approximate time: 12 minutes. Please refer to the DGM models in the Models Booklet (Section B). Each DGM model captures one observable action (e.g., a detected event), the context-awareness evidence that surrounds it, multiple goal hypotheses with conditions, a precedence ordering, response bindings, and an escalation strategy.

B.1 Global ratings on the DGM approach

17. B1-1. Treating an observable detection event (e.g., a person crouching near a vehicle) as something that can serve multiple possible goals depending on context is a realistic representation of how the system actually has to behave.

Anchor: 'Agree' means: yes, in the surveillance system the same detection legitimately maps to different responses depending on time, location, actor role, or other context.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

18. B1-2. The kinds of context-awareness evidence used in the DGM models (sensor readings, temporal observations, identity / role information, spatial information, derived signals) match the kinds of information the system has access to in practice.

Anchor: 'Agree' means: the evidence categories cover what the system actually consumes. 'Disagree' means: the models rely on information the system does not have, or omit information it does have.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

19. B1-3. The precedence orderings encoded in the DGM models (e.g., safety / security goals taking priority over convenience goals) reflect how your team would actually prioritize in operational practice.

Anchor: 'Agree' means: the precedence rules match operational policy. Note any disagreements in the per-goal table in B.2.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

20. B1-4. DGM makes context-dependent goal behaviour explicit in a way that the current specification (existing rules and documentation) does not.

Anchor: 'Agree' means: the multiplicity of possible goals for a single detection, and the rules for resolving among them, are visible in the DGM models in a way they are not in the current rules.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

21. B1-5. The DGM escalation strategy (defaulting to a conservative or safe response, optionally requesting additional evidence, when no goal hypothesis is supported) is appropriate for a surveillance system.

Anchor: 'Agree' means: defaulting to caution is the right behaviour in ambiguous situations. Use B1-6 if you think escalation is over-aggressive.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

22. B1-6. The DGM models would produce too many escalations or false alerts in the operating conditions the system faces.

Anchor: 'Agree' means: the escalation logic is too aggressive and would generate noise. 'Disagree' means: the escalation behaviour is appropriately calibrated. This is a check on B1-5.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

B.2 Per-dynamic-goal coverage

For each dynamic goal in the Models Booklet (Section B), please answer three questions: (i) do the goal hypotheses cover the goals the action can serve, (ii) does the precedence ordering match your operational policy, (iii) do the response bindings match what the system actually does or should do?

23. B2-a. Do the hypotheses cover all relevant goals for the action?

Mark only one oval per row.

	Y	Partial	N
DG-1 WeaponDetected	☐	☐	☐
DG-2 PlateRecognized	☐	☐	☐
DG-3 SuspiciousBehaviourNearVehicle	☐	☐	☐
DG-4 SmokeOrFireDetected	☐	☐	☐
DG-5 PersonInRestrictedZone	☐	☐	☐

24. B2-b. Does the precedence ordering match your operational policy?

Mark only one oval per row.

	Y	Partial	N
DG-1 WeaponDetected	☐	☐	☐
DG-2 PlateRecognized	☐	☐	☐
DG-3 SuspiciousBehaviourNearVehicle	☐	☐	☐
DG-4 SmokeOrFireDetected	☐	☐	☐
DG-5 PersonInRestrictedZone	☐	☐	☐

25. B2-c. Do the response bindings match what the system actually does or should do?

Mark only one oval per row.

	Y	Partial	N
DG-1 WeaponDetected	☐	☐	☐
DG-2 PlateRecognized	☐	☐	☐
DG-3 SuspiciousBehaviourNearVehicle	☐	☐	☐
DG-4 SmokeOrFireDetected	☐	☐	☐
DG-5 PersonInRestrictedZone	☐	☐	☐

26. B2 comments. If you marked 'Partial' or 'N' in any cell above, please specify which dynamic goal and what is missing or wrong (reference the goal by ID, e.g., 'DG-3: ...').

27. B2-1. Are there observable events in the system that should be modelled as dynamic goals (i.e., where the same event legitimately leads to different responses depending on context) but are not represented above? If yes, please name the event and outline the alternative goals.

B.3 Scenario walkthrough — DG-3 SuspiciousBehaviourNearVehicle

Scenario:

It is 14:00 on a weekday. A person is detected crouching next to a parked car in a campus parking lot for approximately 35 seconds. The license plate of the car is registered to a faculty member. Two maintenance staff are visible in the same parking lot, wearing the maintenance uniform recognised by the body-pose model. There is no prior alert about this lot.

Please walk through the DG-3 model with this evidence and answer the questions below.

28. B3-1. Walking through the DG-3 model with the evidence above, the response the model selects is the response you would consider correct operationally.

Anchor: Note in B3-3 if the model's selected response differs from what you would expect.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

29. B3-2. Which goal hypothesis does the DG-3 model select for the scenario above (e.g., RoutineActivityNoAction, SuspiciousActivityAlert, MaintenanceActivityNoAction, EscalateInsufficientEvidence)? Please name it.

30. B3-3. If the model's selected response differs from what you would do operationally, please explain the gap. What evidence or precedence rule would need to change?

Section C. CARDML4IoT Dependency Models

Supports RQ3 (Dependency Clarity). Approximate time: 7 minutes. Please refer to the CARDML4IoT dependency models in the Models Booklet (Section C). These models represent dependencies among context-awareness requirements as either temporal (one must precede another), data (multiple requirements consume the same data source), or resource (multiple requirements compete for the same device or service).

C.1 Global ratings on CARDML4IoT

31. C1-1. The CARDML4IoT models make dependencies among context-awareness requirements visible in a way that the existing system documentation does not.

Anchor: 'Agree' means: I can see, from these models, which CA requirements share devices, share data, or constrain each other's order; in the existing documentation I cannot see this directly.

Mark only one oval.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

32. C1-2. The temporal, data, and resource categories of dependency are sufficient to express the kinds of dependencies that actually occur in the system.

Anchor: 'Agree' means: every dependency I can think of in the system fits one of these three categories.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

33. C1-3. The CARDML4IoT models would help me reason about the impact of adding, removing, or modifying a context-awareness requirement.

Anchor: 'Agree' means: I could use the model to identify what else would be affected by such a change.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

34. C1-4. The CARDML4IoT models add complexity without adding insight beyond what the UCM4IoT use cases already convey.

Anchor: 'Agree' means: CARDML4IoT is redundant given the use cases. 'Disagree' means: it adds something the use cases alone do not give you. This is a check on C1-1.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

35. C2. Are there dependencies among CA requirements that the CARDML4IoT models did not capture, or types of dependency that do not fit the temporal / data / resource categorization? Please describe.

Section D. Comparative Assessment vs Current Approach

Supports RQ4 (Practical Utility). Approximate time: 6 minutes. This section asks you to rate your team's CURRENT specification approach (existing rules, internal documentation, and any goal/i* models you use) and the PROPOSED approach (UCM4IoT + DGM + CARDML4IoT) on the same dimensions, side by side. Please rate each on a 1–5 scale where 1 = very poor and 5 = very good.

36. D1-current. Rate the CURRENT approach (existing rules / docs)

1 = very poor on this dimension, 5 = very good.

Mark only one oval per row.

	1	2	3	4	5
Coverage of functional requirements	☐	☐	☐	☐	☐
Coverage of context-awareness aspects (time, location, role, mode)	☐	☐	☐	☐	☐
Explicit handling of multiple possible goals for a single event	☐	☐	☐	☐	☐
Visibility of dependencies among CA requirements	☐	☐	☐	☐	☐
Traceability between requirements and design / operational rules	☐	☐	☐	☐	☐
Support for impact analysis when a requirement changes	☐	☐	☐	☐	☐
Effort to construct the specification (1 = much	☐	☐	☐	☐	☐

(1, much
effort, 5
little effort)
little effort)

37. D1-proposed. Rate the PROPOSED approach (UCM4IoT + DGM + CARDML4IoT)

1 = very poor on this dimension, 5 = very good.

Mark only one oval per row.

	1	2	3	4	5
Coverage of functional requirements	☐	☐	☐	☐	☐
Coverage of context-awareness aspects (time, location, role, mode)	☐	☐	☐	☐	☐
Explicit handling of multiple possible goals for a single event	☐	☐	☐	☐	☐
Visibility of dependencies among CA requirements	☐	☐	☐	☐	☐
Traceability between requirements and design / operational rules	☐	☐	☐	☐	☐
Support for impact analysis when a requirement changes	☐	☐	☐	☐	☐
Effort to construct the specification (1 = much	☐	☐	☐	☐	☐

(1 = much
effort, 5 =
little effort)
little effort)

38. D1-1. On which dimensions, if any, does the current approach outperform the proposed approach? Please be concrete.

39. D1-2. On which dimensions, if any, does the proposed approach most clearly outperform the current approach?

Section E. Perceived Usefulness, Ease of Use, and Intention to Use

Approximate time: 5 minutes. Please answer based on your overall impression of the proposed approach (UCM4IoT + DGM + CARDML4IoT) as a single combined method.

E.1 Perceived ease of use

40. E1-1. It is easy for me to understand what a model written in this approach is saying about the system.

Anchor: 'Agree' means: reading a model and explaining what it means takes reasonable effort, comparable to reading a use case or a goal model.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

41. E1-2. The proposed approach is cumbersome to use.

Anchor: 'Agree' means: the notation, structure, or amount of detail required gets in the way.
'Disagree' means: it is appropriately economical for what it captures.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

E.2 Perceived usefulness

42. E2-1. Using the proposed approach would improve my team's ability to specify requirements for context-aware IoT systems.

Anchor: 'Agree' means: requirements written in this approach would be more complete, more precise, or more traceable than what your team produces today.

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

43. E2-2. The benefits of using the proposed approach outweigh the additional modelling effort it requires.

Anchor: 'Agree' means: the time invested in producing the models pays off in clearer specifications, fewer downstream surprises, or better maintenance.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

E.3 Intention to use

44. E3-1. I would consider using the proposed approach (or a similar combined approach) in future projects on context-aware IoT systems.

Anchor: 'Agree' means: I would seriously evaluate adopting this approach for a future project.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

45. E3-2. I would prefer to continue using my team's current approach rather than switch to the proposed approach.

Anchor: 'Agree' means: I do not see a reason to switch from current practice. This is a check on E3-1.

Mark only one oval.

	1	2	3	4	5	
Stro	☐	☐	☐	☐	☐	Strongly Agree

Section F. Open-Ended Overall Feedback

Approximate time: 2 minutes. Short answers are completely fine.

46. F-1. What do you see as the main STRENGTHS of the proposed approach (UCM4IoT + DGM + CARDML4IoT) for a system like the University Surveillance System?

47. F-2. What do you see as the main WEAKNESSES or limitations of the proposed approach? If you would change one thing about the notation, process, or tooling, what would it be?

48. F-3. Any additional comments, suggestions, or observations?

Google Forms

